My teaching philosophy is informed by the understanding that learning involves the construction and refinement of mental frameworks—schemas—that allow students to organize, interpret, and apply new information. Drawing on cognitive science and schema theory, I think of instruction as a collaborative process that helps students build conceptual models that connect new knowledge to existing understanding. Students arrive in the classroom with varying levels of prior knowledge, existing schemas, and misconceptions developed through previous experiences. Schema theory suggests that learning occurs through three primary mechanisms: accretion (adding new information to existing schemas), tuning (refining schemas based on new experiences), and restructuring (fundamentally reorganizing schemas when new information challenges existing frameworks). My work as an instructor involves facilitating these processes in ways that support students' individual learning needs.

While traditional instruction often presents concepts in a top-down manner—providing frameworks before students engage with examples—I have found value in a bottom-up approach. This means beginning with concrete examples, observations, and guided exploration before working together to abstract general principles. In biology courses, students might examine multiple specific physiological processes before identifying underlying patterns and core schemas that unite these examples. Research suggests that when students actively participate in constructing their understanding through discussion and analysis of examples, they may develop more flexible schemas than when frameworks are simply presented to them. Group discussions and case studies become useful tools, allowing students to articulate their developing understanding and refine their mental models through interaction. My instructional design attempts to link new information to existing knowledge. I try to begin units by assessing students' prior knowledge—not to judge, but to understand the schemas they bring and identify potential misconceptions. Throughout instruction, I work to make connections between new concepts and material students have previously encountered, helping them extend their existing schemas rather than creating isolated pockets of knowledge.

When introducing complex biological systems, I find it helpful to scaffold learning by first ensuring students have foundational concepts in place. This layered approach may reduce cognitive load and allow students to focus on integrating new information rather than struggling with prerequisite knowledge. Misconceptions represent incompletely formed or inaccurate schemas, and addressing them seems to require more than simply presenting correct information. I try to incorporate opportunities for reflection, asking students to examine their assumptions and consider where their existing understanding might be incomplete. Through feedback on assignments, I hope to help students recognize patterns in their errors and understand not just what they got wrong, but how to restructure their thinking.

Students appear to construct schemas at different rates and through different pathways. Some may need more concrete examples before abstracting principles; others may benefit from seeing organizational frameworks earlier. By incorporating varied instructional methods—recorded lectures for flexible review, in-class discussions for collaborative learning, case studies for application, and visual representations for conceptual organization—I try to provide multiple entry points for schema construction. My assessment strategies aim to evaluate not only recall of facts but also the quality and flexibility of students' schemas. Rather than focusing exclusively on multiple-choice questions, I tend to emphasize short and long response questions that ask students to demonstrate conceptual understanding, make connections across topics, and apply schemas to novel situations. This approach also provides information about how students are organizing knowledge and where their schemas might need refinement.

Effective teaching seems to require understanding not just what we teach, but how students learn. By grounding my instruction in schema theory and cognitive science, I work to create learning experiences that help students build interconnected knowledge structures. My goal is to help students develop conceptual frameworks that will allow them to continue learning and applying their knowledge beyond the classroom. This commitment to schema-based instruction shapes my approach to teaching, from lesson planning to assessment design, and reflects a belief that deep understanding develops from thoughtfully constructed mental models rather than memorized facts.